

Only solving 1(a) here because the other problems are almost identical.

Problem 1

Solution:

(a) Let $s_n < A < S_n$

s_n is the lower bound

S_n is the upper bound

A is the actual area under the curve over the interval $[0, b]$.

Then,

$$\begin{aligned}
 s_n &= \left(\frac{b}{n}\right) \left(2 \left(\frac{0b}{n}\right)^2\right) + \left(\frac{b}{n}\right) \left(2 \left(\frac{1b}{n}\right)^2\right) + \cdots + \left(\frac{b}{n}\right) \left(2 \left(\frac{(n-1)b}{n}\right)^2\right) \\
 &= 2 \left(\frac{b^3}{n^3}\right) [1^2 + \cdots + (n-1)^2] \\
 &= 2 \left(\frac{b^3}{n^3}\right) \left[\left(\frac{(n-1)^3}{3} + \frac{(n-1)^2}{2} + \frac{(n-1)}{6}\right)\right] \\
 &= 2 \left[\left(\frac{b^3}{3} - \frac{b^3}{2n} + \frac{b^3}{6n^2}\right)\right] \\
 &= \left(\frac{2b^3}{3}\right) - \frac{b^3}{n} + \frac{b^3}{3n^2} \\
 &= \left(\frac{2b^3}{3}\right) + \left(\frac{b^3}{n}\right) \left(-1 + \frac{1}{3n}\right)
 \end{aligned}$$

and

$$\begin{aligned}
 S_n &= \left(\frac{b}{n}\right) \left(2 \left(\frac{1b}{n}\right)^2\right) + \left(\frac{b}{n}\right) \left(2 \left(\frac{2b}{n}\right)^2\right) + \cdots + \left(\frac{b}{n}\right) \left(2 \left(\frac{nb}{n}\right)^2\right) \\
 &= 2 \left(\frac{b^3}{n^3}\right) [1^2 + \cdots + n^2] \\
 &= 2 \left(\frac{b^3}{n^3}\right) \left[\left(\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}\right)\right] \\
 &= 2 \left[\left(\frac{b^3}{3} + \frac{b^3}{2n} + \frac{b^3}{6n^2}\right)\right] \\
 &= \left(\frac{2b^3}{3}\right) + \frac{b^3}{n} + \frac{b^3}{3n^2} \\
 &= \left(\frac{2b^3}{3}\right) + \left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right)
 \end{aligned}$$

Now, observe that for any $n > 0$

$$\left(\frac{b^3}{n}\right) \left(-1 + \frac{1}{3n}\right) < 0$$

Thus,

$$s_n = \left(\frac{2b^3}{3}\right) + \left(\frac{b^3}{n}\right) \left(-1 + \frac{1}{3n}\right) < \left(\frac{2b^3}{3}\right) + 0 = \left(\frac{2b^3}{3}\right)$$

Conversely, for all $n > 0$

$$\left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right) > 0$$

so we have

$$S_n = \left(\frac{2b^3}{3}\right) + \left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right) > \left(\frac{2b^3}{3}\right) + 0 = \left(\frac{2b^3}{3}\right)$$

Hence,

$$s_n < A = \left(\frac{2b^3}{3}\right) < S_n$$

is possible. To show that this is the *only* possible value for A for all $n > 0$ we must show

that neither $A < \left(\frac{2b^3}{3}\right)$ nor $A > \left(\frac{2b^3}{3}\right)$ work for all $n > 0$

For the former, let $A = \left(\frac{2b^3}{3}\right) - c$ where $c > 0$. If we can show that there exists $n > 0$ such that $s_n > A$, then we have a contradiction, because we know that $s_n < A < S_n$ must be true. Said another way, we must show that $s_n - A > 0$. Now,

$$\begin{aligned} s_n - A &= \left[\left(\frac{2b^3}{3}\right) + \left(\frac{b^3}{n}\right) \left(-1 + \frac{1}{3n}\right) \right] - \left[\left(\frac{2b^3}{3}\right) - c \right] \\ &= \left(\frac{b^3}{n}\right) \left(-1 + \frac{1}{3n}\right) + c \end{aligned}$$

Here, we know that the first part is always < 0 for any $n > 0$, but observe that if $n \gg 0$, it's just *slightly* negative. And as we keep increasing n it gets closer and closer to 0 (i.e. less and less negative). As a result, no matter how small c is, there will always be some n such that

$$0 > \left(\frac{b^3}{n}\right) \left(-1 + \frac{1}{3n}\right) > -c$$

which means that

$$\begin{aligned} s_n - A &= \left(\frac{b^3}{n}\right) \left(-1 + \frac{1}{3n}\right) + c \\ &> -c + c \\ &= 0 \end{aligned}$$

Thus, $A < \left(\frac{2b^3}{3}\right)$ doesn't work.

Now, for the other case, let $A = \left(\frac{2b^3}{3}\right) + c > \left(\frac{2b^3}{3}\right)$ where $c > 0$. If we can show that there exists $n > 0$ such that $S_n < A$ then we have a contradiction, because we know that $s_n < A < S_n$ must be true. Said another way, we must show that $A - S_n > 0$.

$$\begin{aligned} A - S_n &= \left[\left(\frac{2b^3}{3}\right) + c\right] - \left[\left(\frac{2b^3}{3}\right) + \left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right)\right] \\ &= c - \left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right) \end{aligned}$$

Here, we know that the second operand in the subtraction is always > 0 for any $n > 0$, but observe that if $n \gg 0$, it's just *slightly* positive. And as we keep increasing n it gets closer and closer to 0 (i.e. less and less positive). As a result, no matter how small c is, there will always be some n such that

$$0 < \left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right) < c$$

which means that

$$-c > -\left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right) > 0$$

and therefore

$$\begin{aligned} A - S_n &= c - \left(\frac{b^3}{n}\right) \left(1 + \frac{1}{3n}\right) \\ &> c - c \\ &= 0 \end{aligned}$$

as desired. We've thus shown that $A = \frac{2b^3}{3}$ is the only option that works.

□